

Savannah L. Ferretti

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EDUCATION

Ph.D. in Earth System Science , University of California, Irvine	Jun 2026
M.S. in Earth System Science , University of California, Irvine	Dec 2023
B.S. in Earth & Atmospheric Sciences , Cornell University	Dec 2020

EXPERIENCE

Graduate Student Researcher Sep 2021–Jun 2026
Pritchard & Baldwin Labs, University of California, Irvine Irvine, CA

- Used data-driven and physics-based approaches to identify & explain drivers of South Asian monsoon rainfall variability
- Developed an interpretable machine learning methodology that used neural networks & equation discovery to learn physically meaningful relationships from high-dimensional atmospheric data
- Built open-source Python pipelines on GitHub for cloud data access, large-array processing, uncertainty analysis, model evaluation, & visualization in HPC environments
- Communicated findings through peer-reviewed publications & conference presentations

Consulting Analyst Mar–Aug 2021
Higher Education Research Services, Huron Consulting Group New York, NY

- Built Excel-based financial planning & reporting tools for a research nonprofit
- Developed long-term resource forecasts & presented recommendations to executive stakeholders

Undergraduate Research Assistant Jun–Aug 2020
Commene Lab, Lamont-Doherty Earth Observatory Palisades, NY

- Quantified changes in atmospheric methane concentrations during the NYC COVID-19 shutdown
- Applied meteorological analysis & transport modeling to attribute emissions to likely source regions

Student Administrative Assistant Jan–Dec 2020
CALS Office of Admissions, Cornell University Ithaca, NY

- Increased prospective student yield through creation of an interactive [ArcGIS campus tour](#)
- Improved student engagement through execution of an office-wide social media strategy

Undergraduate Research Assistant Sep 2019–May 2020
Pritchard Lab, Cornell University Ithaca, NY

- Processed satellite imagery to construct time series of volcanic thermal anomalies
- Investigated pre-eruptive thermal signals for inclusion in NASA's AVTOD volcano database

SKILLS

Python (PyTorch, PySR, Dask, Xarray, Pandas, Cartopy, Matplotlib) • Bash • Git/GitHub • High-Performance/Distributed Computing • NetCDF/HDF5/Zarr • Linux • Microsoft Excel ([Certified Specialist](#)) • $\text{\LaTeX}$ • WordPress

PUBLICATIONS

- S. L. Ferretti**, J. W. Baldwin, M. S. Pritchard, & T. Beucler. Data-driven equation discovery of thermodynamic–precipitation relationships for the South Asian monsoon. In preparation for *Journal of Advances in Modeling Earth Systems*.
- S. L. Ferretti**, J. Lin, S. Shamekh, J. W. Baldwin, M. S. Pritchard, & T. Beucler. (2026). [Data-driven integration kernels for interpretable nonlocal operator learning](#). In press at *Environmental Data Science*.
- S. L. Ferretti**, M. S. Pritchard, F. Ahmed, L. Peng, & J. W. Baldwin. (2025). [Explaining South Asian monsoon rainfall seasonality using a metric of plume buoyancy](#). *Geophysical Research Letters*, 52(16), e2025GL115546.
- L. Peng, P. N. Blossey, W. M. Hannah, ...**S. L. Ferretti**, ...& M. S. Pritchard. (2025). [Resolving low cloud feedbacks globally with E3SM High-Res MMF: Agreement with LES but stronger shortwave effects](#). *Journal of Advances in Modeling Earth Systems*, 17(6), e2025MS005003.
- S. Yu, Z. Hu, A. Subramaniam, ...**S. L. Ferretti**, ...& M. S. Pritchard. (2025). [ClimSim-Online: A large multi-scale dataset and framework for hybrid ML–physics climate emulation](#). *Journal of Machine Learning Research*. 26(142), 1-85.
- S. Yu, W. M. Hannah, L. Peng, ...**S. L. Ferretti**, ...& M. S. Pritchard. (2023). [ClimSim: A large multi-scale dataset for hybrid physics–ML climate emulation](#). *NeurIPS 2023 Datasets & Benchmarks Track*, **Outstanding Paper Award**.
- A. M. Jenney, **S. L. Ferretti**, & M. S. Pritchard. (2023). [Vertical resolution impacts explicit simulation of deep convection](#). *Journal of Advances in Modeling Earth Systems*, 15(10), e2022MS003444.

PRESENTATIONS

- (Oral) **S. L. Ferretti**, J. Lin, S. Shamekh, ...& T. Beucler (2026). Data-Driven Integration Kernels for Interpretable Nonlocal Operator Learning. *Climate Informatics 2026*. Lausanne, VD, CH.
- (Invited Oral) **S. L. Ferretti**, J. Lin, S. Shamekh, ...& T. Beucler (2025). Making Data-Driven Climate Prediction Interpretable by Design. *Learning the Earth with Artificial Intelligence and Physics Lectures in Climate Data Science*. Virtual.
- (Oral) **S. L. Ferretti**, T. Beucler, M. S. Pritchard, ...& J. W. Baldwin (2025). Learning Nonlocal Controls on South Asian Monsoon Rainfall. *American Geophysical Union Fall Meeting*. New Orleans, LA, USA.
- (Poster) **S. L. Ferretti**, M. S. Pritchard, & J. W. Baldwin (2025). Towards Data-Driven Discovery of Thermodynamic Controls on South Asian Monsoon Rainfall. *Learning the Earth with Artificial Intelligence and Physics Year 5 Meeting*. Briarcliff Manor, NY, USA.
- (Poster) **S. L. Ferretti**, L. Peng, J. W. Baldwin, & M. S. Pritchard (2024). Spatiotemporal Stress-Testing of a Process-Oriented Diagnostic for Rainfall. *American Geophysical Union Fall Meeting*. Washington DC, USA.
- (Poster) **S. L. Ferretti**, J. W. Baldwin, N. Liu, ...& M. S. Pritchard (2024). Unraveling CMIP6 Biases in Summer Monsoon Rainfall Over the Arabian Sea and Western India. *104th American Meteorological Society Annual Meeting*. Baltimore, MD, USA.
- (Poster) **S. L. Ferretti**, N. Liu, J. W. Baldwin, & M. S. Pritchard (2022). Understanding the Monthly Variation in the Upstream Enhancement of Indian Summer Monsoon Precipitation Near the Western Ghats. *American Geophysical Union Fall Meeting*. Chicago, IL, USA.

(Invited Oral) **S. L. Ferretti** (2022). Using Models to Understand Climate Change. *Making Modern Science*, University of California, Irvine. Irvine, CA, USA.

(Poster) **S. L. Ferretti**, B. Dalton, L. D. Schiferl, ...& R. Commene (2020). Assessing the Meteorological Impact on Methane (CH₄) Emission Changes in New York City During the COVID-19 Shutdown. *American Geophysical Union Fall Meeting*. Virtual.

AWARDS & FELLOWSHIPS

Advancing Inclusive Excellence Award School of Physical Sciences, University of California, Irvine	Jan–Jun 2023 \$ 400
Diversity, Equity, & Inclusion Graduate Leaders Fellowship School of Physical Sciences, University of California, Irvine	Sep 2022–Jun 2023 \$ 24,720
Class Award Cornell University Class Council	Apr 2018

TEACHING

University of California, Irvine , Introduction to Earth System Science Teaching Assistant	Jun–Jul 2026 Virtual
University of California, Irvine , Air Quality Management Teaching Assistant	Apr–Jun 2024 Irvine, CA
University of California, Irvine , Earth System Chemistry Teaching Assistant	Sep–Dec 2023 Irvine, CA
University of California, Irvine , Diversity in STEM Seminar Coordinator	Sep–Dec 2022 Irvine, CA

SERVICE & ENGAGEMENT

Journal of Geophysical Research Atmospheres , Reviewer	Nov 2025–Present
UCI ESS First-Year Mentoring Program , Mentor	Sep 2022–Aug 2024
UCI Community Geoscience Initiative , Fellowship Application Reviewer	Jun 2023
UCI CLEWS Climate-Tech Seminar Series , Panelist	Jun 2023
UCI ESS Inclusive Excellence Committee , Organizer	Sep 2022–Jun 2023
NCAR Community Earth System Model Tutorial , Participant	Aug 2022
AGU Bridge Program New Student Orientation , Panelist	Jul 2022
Seed Consultant Group , Volunteer Project Consultant	Mar–May 2022
GREEN Program Nepal , Microgrid Systems for Rural Development, Participant	Jan 2020
Cornell University Dance Team , Cofounder & Vice President	Oct 2017–Dec 2020
Cornell University College Mentors for Kids , Mentor	Aug 2017–May 2018

RELEVANT COURSEWORK _____

GRADUATE

Machine Learning for Environmental Engineering & Science • Atmospheric Dynamics • Geoscience Modeling & Data Analysis • Environmental Fluid Mechanics & Turbulence • Global Physical Climatology • Humans in the Earth System • Planning for Climate Justice • Communication Skills for Environmental Scholars

UNDERGRADUATE

Atmospheric Thermodynamics & Hydrostatics • Climate Dynamics • Microclimatology • Meteorological Observations & Instrumentation • Synoptic Meteorology • Evolution of the Earth System • Interior of the Earth • Oral Communication • Leading People & Effective Communication